POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5014D | Semester V | Theory | 4 Credits

Building Maintenance & Services

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Civil Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5014D as Building Maintenance & Services in Semester V for Civil Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Delhi’s Fire Protection, Services and Maintenance course, the Swayam/IGNOU course on Repair and Maintenance of Buildings and the MSBTE polytechnic syllabus. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Building repairs, structural retrofitting, electrical wiring, plumbing layouts and fire-protection systems must be based on approved engineering designs, authorised safety inspections, current building codes and competent professional review.

Verified source note: Verified sources support maintenance management, building defects, deterioration, repair, water/drainage services, electrical/lighting and HVAC/mechanical services. The four-module sequence is a learning sequence, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Building Maintenance & Services studies the systematic upkeep, repair and management of building systems to ensure safety, efficiency and durability. It develops the ability to manage maintenance planning, identify and remediate building defects, design water and drainage layouts, coordinate electrical and lighting services, and understand HVAC and fire protection while respecting the technical and safety boundaries of building operations.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Building construction, construction materials, basic hydraulics, elementary electrical awareness and safe site practice.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain maintenance management principles, planning, records and the role of the maintenance engineer.
2. Explain building defects, causes of deterioration and methods for repair and retrofitting.
3. Explain water supply and drainage services, including plumbing fixtures, layouts and maintenance.
4. Explain electrical, lighting, HVAC and mechanical services, including safety and fire protection.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain maintenance types, planning procedures and the management of building records.	Understand
CO2	Explain building defects, deterioration mechanisms and the principles of effective repair.	Understand
CO3	Explain water supply and drainage systems, plumbing fixtures and their maintenance needs.	Understand
CO4	Explain electrical, lighting, HVAC and fire-protection services within building operations.	Understand

Official CO–PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Maintenance Management and Building Records	Importance of maintenance; types (preventive, corrective, routine); maintenance planning and scheduling; maintenance organization and records; role of the maintenance engineer.	Approx. 14 hours	CO1	Module 1
2	Building Defects, Deterioration and Repairs	Defects in buildings (cracks, dampness, efflorescence); causes of deterioration (environmental, workmanship, material); methods of repair and retrofitting; moisture control and waterproofing.	Approx. 16 hours	CO2	Module 2
3	Water Supply and Drainage Services	Water supply systems in buildings; plumbing fixtures and fittings; hot and cold water distribution; drainage and sewerage systems; traps and vents; maintenance of plumbing.	Approx. 16 hours	CO3	Module 3
4	Electrical, HVAC and Mechanical Services	Electrical wiring and safety; lighting and lamps; HVAC basics (ventilation, air conditioning); lifts and escalators; fire protection and safety systems; energy conservation in services.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Maintenance Management and Building Records

Importance of maintenance; types (preventive, corrective, routine); maintenance planning and scheduling; maintenance organization and records; role of the maintenance engineer.

CO1 · Approx. 14 hours

Building Defects, Deterioration and Repairs

Defects in buildings (cracks, dampness, efflorescence); causes of deterioration (environmental, workmanship, material); methods of repair and retrofitting; moisture control and waterproofing.

CO2 · Approx. 16 hours

Water Supply and Drainage Services

Water supply systems in buildings; plumbing fixtures and fittings; hot and cold water distribution; drainage and sewerage systems; traps and vents; maintenance of plumbing.

CO3 · Approx. 16 hours

Electrical, HVAC and Mechanical Services

Electrical wiring and safety; lighting and lamps; HVAC basics (ventilation, air conditioning); lifts and escalators; fire protection and safety systems; energy conservation in services.

CO4 · Approx. 14 hours

MODULE 1

Maintenance Management and Building Records

Importance of maintenance; types (preventive, corrective, routine); maintenance planning and scheduling; maintenance organization and records; role of the maintenance engineer.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

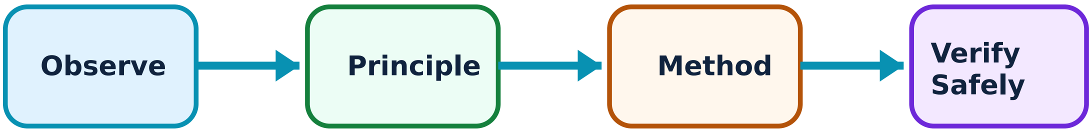
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Maintenance Management and Building Records: identify the system, state its principle, follow the correct method and verify the result safely.

1. Maintenance types and planning−

Maintenance ensures buildings remain functional and safe. Preventive maintenance (scheduled) aims to prevent failure, while corrective maintenance (breakdown) fixes issues after they occur. Planning involves scheduling inspections, allocating resources and managing work orders.

Study and application method

List four objectives of building maintenance and create a conceptual annual maintenance schedule for a small school building.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List four objectives of building maintenance and create a conceptual annual maintenance schedule for a small school building.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Deferred maintenance leads to higher costs and safety risks. Always prioritise safety-critical repairs and document all maintenance activities.

2. Maintenance organization and records−

Effective maintenance requires an organized team and detailed records. Records include equipment manuals, inspection reports, complaint registers, history cards and cost data, which inform future maintenance decisions and budgeting.

Study and application method

Identify the essential records for a building's water supply system and draft a simple complaint-register format.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Missing or inaccurate records hinder maintenance efficiency and can lead to safety lapses. Maintain all building data in an authorised, accessible system.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Building Defects, Deterioration and Repairs

Defects in buildings (cracks, dampness, efflorescence); causes of deterioration (environmental, workmanship, material); methods of repair and retrofitting; moisture control and waterproofing.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

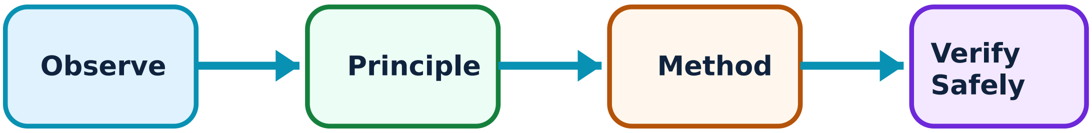
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Building Defects, Deterioration and Repairs: identify the system, state its principle, follow the correct method and verify the result safely.

1. Deterioration and common defects−

Buildings deteriorate due to weather, chemical action, structural movement and poor workmanship. Common defects include dampness (seepage), various types of cracks (structural, thermal), efflorescence and corrosion of reinforcement.

Study and application method

Sketch three types of cracks in a building and explain the likely cause and a conceptual repair method for each.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch three types of cracks in a building and explain the likely cause and a conceptual repair method for each.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Surface repairs may hide underlying structural issues. Always identify the root cause of a defect before performing repairs.

2. Repair methods and retrofitting−

Repairs restore functionality and aesthetics. Techniques include grouting, guniting, epoxy injection and waterproofing. Retrofitting strengthens existing structures to meet new requirements or improve seismic resistance, requiring careful structural assessment.

Study and application method

Compare the principles of grouting and guniting for repairing concrete structures; list three methods for basement waterproofing.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the principles of grouting and guniting for repairing concrete structures; list three methods for basement waterproofing.

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Common mistake / precaution: Structural repairs and retrofitting must be designed and supervised by a competent structural engineer. Unauthorised structural changes can lead to collapse.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Water Supply and Drainage Services

Water supply systems in buildings; plumbing fixtures and fittings; hot and cold water distribution; drainage and sewerage systems; traps and vents; maintenance of plumbing.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

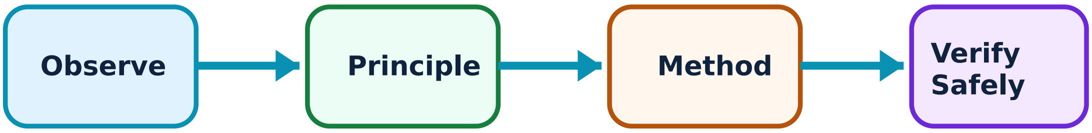
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Water Supply and Drainage Services: identify the system, state its principle, follow the correct method and verify the result safely.

1. Water supply and plumbing fixtures−

Building water services provide safe water for various uses. Systems include storage tanks, pumps and distribution piping. Fixtures (taps, valves) and fittings must be selected for durability and ease of maintenance to prevent leaks and waste.

Study and application method

Draw a conceptual layout for a residential water supply system including an overhead tank and identify three types of plumbing valves.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a conceptual layout for a residential water supply system including an overhead tank and identify three types of plumbing valves.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Water systems must prevent backflow and contamination. All plumbing work must comply with local public health and plumbing codes.

2. Drainage, traps and venting−

Drainage systems remove waste and rainwater. Traps (P, Q, S) prevent sewer gases from entering buildings, while vents ensure air circulation and prevent siphonage. Regular cleaning of gully traps and manholes is essential for system hygiene.

Study and application method

Sketch a P-trap and explain how it prevents the entry of sewer gases; describe the difference between a single-stack and a two-pipe drainage system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a P-trap and explain how it prevents the entry of sewer gases; describe the difference between a single-stack and a two-pipe drainage system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Improper drainage leads to foul odours and health hazards. Do not bypass traps or block vents, and ensure all waste is discharged into authorised sewer systems.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Electrical, HVAC and Mechanical Services

Electrical wiring and safety; lighting and lamps; HVAC basics (ventilation, air conditioning); lifts and escalators; fire protection and safety systems; energy conservation in services.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

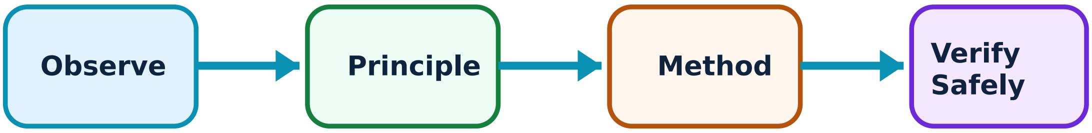
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Electrical, HVAC and Mechanical Services: identify the system, state its principle, follow the correct method and verify the result safely.

1. Electrical and lighting services−

Electrical services provide power and light. Systems include distribution boards, wiring, protection devices (MCBs, ELCBs) and various lamps. Safety involves proper grounding, load management and regular inspection of insulation and connections.

Study and application method

List four electrical safety devices used in buildings and explain the importance of an Earth Leakage Circuit Breaker (ELCB).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List four electrical safety devices used in buildings and explain the importance of an Earth Leakage Circuit Breaker (ELCB).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Electrical faults are a major cause of building fires. All work must be performed by certified electricians according to authorised electrical codes.

2. HVAC, lifts and fire protection−

HVAC ensures indoor comfort and air quality. Lifts and escalators provide vertical transport. Fire protection includes detection (alarms) and suppression (sprinklers, extinguishers). All mechanical services require regular testing and certification for safe operation.

Study and application method

Identify the components of a building's fire-alarm system and describe the maintenance needs for a passenger lift.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify the components of a building's fire-alarm system and describe the maintenance needs for a passenger lift.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Fire-safety and lift systems are life-critical. Do not tamper with alarms or sprinklers, and ensure all mechanical services are maintained by authorised service providers.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Maintenance Management and Building Records

Use the concept flow to revise observation, principle, method and verification.

Module 2: Building Defects, Deterioration and Repairs

Use the concept flow to revise observation, principle, method and verification.

Module 3: Water Supply and Drainage Services

Use the concept flow to revise observation, principle, method and verification.

Module 4: Electrical, HVAC and Mechanical Services

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare a preventive-maintenance checklist for a building's plumbing and electrical systems.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch three common building defects and describe their causes and conceptual repairs.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a layout for a simple residential drainage system showing traps and vents.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

List the electrical safety devices and fire-protection equipment required for a high-rise building.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Design a simple record-keeping system for building maintenance complaints and actions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Maintenance Management and Building Records

Explain Maintenance types and planning and state one correct method, application or precaution.−

Maintenance ensures buildings remain functional and safe. Preventive maintenance (scheduled) aims to prevent failure, while corrective maintenance (breakdown) fixes issues after they occur. Planning involves scheduling inspections, allocating resources and managing work orders.

Answer framework: List four objectives of building maintenance and create a conceptual annual maintenance schedule for a small school building.

Important point: Deferred maintenance leads to higher costs and safety risks. Always prioritise safety-critical repairs and document all maintenance activities.

Explain Maintenance organization and records and state one correct method, application or precaution.−

Effective maintenance requires an organized team and detailed records. Records include equipment manuals, inspection reports, complaint registers, history cards and cost data, which inform future maintenance decisions and budgeting.

Answer framework: Identify the essential records for a building's water supply system and draft a simple complaint-register format.

Important point: Missing or inaccurate records hinder maintenance efficiency and can lead to safety lapses. Maintain all building data in an authorised, accessible system.

Module 2 — Building Defects, Deterioration and Repairs

Explain Deterioration and common defects and state one correct method, application or precaution.−

Buildings deteriorate due to weather, chemical action, structural movement and poor workmanship. Common defects include dampness (seepage), various types of cracks (structural, thermal), efflorescence and corrosion of reinforcement.

Answer framework: Sketch three types of cracks in a building and explain the likely cause and a conceptual repair method for each.

Important point: Surface repairs may hide underlying structural issues. Always identify the root cause of a defect before performing repairs.

Explain Repair methods and retrofitting and state one correct method, application or precaution.−

Repairs restore functionality and aesthetics. Techniques include grouting, guniting, epoxy injection and waterproofing. Retrofitting strengthens existing structures to meet new requirements or improve seismic resistance, requiring careful structural assessment.

Answer framework: Compare the principles of grouting and guniting for repairing concrete structures; list three methods for basement waterproofing.

Important point: Structural repairs and retrofitting must be designed and supervised by a competent structural engineer. Unauthorised structural changes can lead to collapse.

Module 3 — Water Supply and Drainage Services

Explain Water supply and plumbing fixtures and state one correct method, application or precaution.−

Building water services provide safe water for various uses. Systems include storage tanks, pumps and distribution piping. Fixtures (taps, valves) and fittings must be selected for durability and ease of maintenance to prevent leaks and waste.

Answer framework: Draw a conceptual layout for a residential water supply system including an overhead tank and identify three types of plumbing valves.

Important point: Water systems must prevent backflow and contamination. All plumbing work must comply with local public health and plumbing codes.

Explain Drainage, traps and venting and state one correct method, application or precaution.−

Drainage systems remove waste and rainwater. Traps (P, Q, S) prevent sewer gases from entering buildings, while vents ensure air circulation and prevent siphonage. Regular cleaning of gully traps and manholes is essential for system hygiene.

Answer framework: Sketch a P-trap and explain how it prevents the entry of sewer gases; describe the difference between a single-stack and a two-pipe drainage system.

Important point: Improper drainage leads to foul odours and health hazards. Do not bypass traps or block vents, and ensure all waste is discharged into authorised sewer systems.

Module 4 — Electrical, HVAC and Mechanical Services

Explain Electrical and lighting services and state one correct method, application or precaution.−

Electrical services provide power and light. Systems include distribution boards, wiring, protection devices (MCBs, ELCBs) and various lamps. Safety involves proper grounding, load management and regular inspection of insulation and connections.

Answer framework: List four electrical safety devices used in buildings and explain the importance of an Earth Leakage Circuit Breaker (ELCB).

Important point: Electrical faults are a major cause of building fires. All work must be performed by certified electricians according to authorised electrical codes.

Explain HVAC, lifts and fire protection and state one correct method, application or precaution.−

HVAC ensures indoor comfort and air quality. Lifts and escalators provide vertical transport. Fire protection includes detection (alarms) and suppression (sprinklers, extinguishers). All mechanical services require regular testing and certification for safe operation.

Answer framework: Identify the components of a building's fire-alarm system and describe the maintenance needs for a passenger lift.

Important point: Fire-safety and lift systems are life-critical. Do not tamper with alarms or sprinklers, and ensure all mechanical services are maintained by authorised service providers.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Maintenance Management and Building Records.

2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Building Defects, Deterioration and Repairs.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Water Supply and Drainage Services.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Electrical, HVAC and Mechanical Services.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Importance of maintenance; types (preventive, corrective, routine); maintenance planning and scheduling; maintenance organization and records; role of the maintenance engineer..
2. Develop a complete answer or practical plan for: Defects in buildings (cracks, dampness, efflorescence); causes of deterioration (environmental, workmanship, material); methods of repair and retrofitting; moisture control and waterproofing..
3. Develop a complete answer or practical plan for: Water supply systems in buildings; plumbing fixtures and fittings; hot and cold water distribution; drainage and sewerage systems; traps and vents; maintenance of plumbing..
4. Develop a complete answer or practical plan for: Electrical wiring and safety; lighting and lamps; HVAC basics (ventilation, air conditioning); lifts and escalators; fire protection and safety systems; energy conservation in services..

LAST-MILE REVISION

Quick Revision

Module 1: Maintenance Management and Building Records

Importance of maintenance; types (preventive, corrective, routine); maintenance planning and scheduling; maintenance organization and records; role of the maintenance engineer.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Building Defects, Deterioration and Repairs

Defects in buildings (cracks, dampness, efflorescence); causes of deterioration (environmental, workmanship, material); methods of repair and retrofitting; moisture control and waterproofing.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Water Supply and Drainage Services

Water supply systems in buildings; plumbing fixtures and fittings; hot and cold water distribution; drainage and sewerage systems; traps and vents; maintenance of plumbing.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Electrical, HVAC and Mechanical Services

Electrical wiring and safety; lighting and lamps; HVAC basics (ventilation, air conditioning); lifts and escalators; fire protection and safety systems; energy conservation in services.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Maintenance types and planning	Maintenance ensures buildings remain functional and safe.
Maintenance organization and records	Effective maintenance requires an organized team and detailed records.
Deterioration and common defects	Buildings deteriorate due to weather, chemical action, structural movement and poor workmanship.
Repair methods and retrofitting	Repairs restore functionality and aesthetics.
Water supply and plumbing fixtures	Building water services provide safe water for various uses.
Drainage, traps and venting	Drainage systems remove waste and rainwater.
Electrical and lighting services	Electrical services provide power and light.
HVAC, lifts and fire protection	HVAC ensures indoor comfort and air quality.

LEARNING RESOURCES

References

Recommended building-maintenance references

1. B. S. Nayak, A Manual on Maintenance of Buildings.
2. P. S. Gahlot and Sanjay Sharma, Building Services.
3. S. M. Patil, Building Maintenance.
4. Current National Building Code (NBC) and local authority regulations.

Approved public building-maintenance sources

- https://nptel.ac.in/courses/105102176
- https://onlinecourses.swayam2.ac.in/nou23_ce03/preview

These sources provide the verified study basis while official Course 5014D detail is unavailable; they do not replace official building codes, authorised structural reports, certified electrical/plumbing designs or authorised safety procedures.